BODY—COSMETIC CRACK IN OUTER DOOR PANEL—LOWER REAR CORNER OF WINDOW OPENING—INSPECTION AND REPAIR PROCEDURE

Article No. 01-18-2

FORD: 1997-1998 F-250 LD
1997-2000 F-150

This article is being republished in its entirety to update the Service Procedure and to include the use of Special Tools.

ISSUE

Some vehicles may exhibit a cosmetic crack in the outer door panel at the lower rear corner of the window opening. This may be caused by excessive stress on the outer door panel.

ACTION

Inspect, repair, or replace outer door panel. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE
READ THIS REPAIR PROCEDURE COMPLETELY BEFORE PERFORMING ANY SERVICE ACTIONS.

NOTE
ALWAYS PERFORM THE DRIVER'S SIDE DOOR REPAIR FIRST!

Inspect both front doors on vehicle to determine which Service Procedure to follow:

- Procedure "A" - Door Crack Repair (less than 51mm/2″ in length)
- Procedure "B" - Crack In Door Panel (longer than 51mm/2″ in length) (Requires outer door panel replacement)

Refer to the Parts listing tables later in this TSB for parts required for each procedure.

NOTE
THE USE OF FORD OR MOTORCRAFT BRAND PRODUCTS ARE MANDATORY ON ALL REPAIR CLAIMS PAID BY FORD UNLESS OTHERWISE SPECIFIED IN EMISSIONS, SAFETY RECALLS, OWNER NOTIFICATION PROGRAMS, TSBs OR OTHER COMPANY PUBLICATIONS. THE COST OF NON-FORD/MOTORCRAFT PRODUCTS USED FOR A WARRANTY OR ESP/ESC, OR AWA REPAIR WITHOUT JUSTIFYING THEIR USE, (I.E., EMERGENCY REPAIR) IS NOT REIMBURSABLE.

Examples of Ford products are: Adhesives, sealers, solvents, cleaners, washes, and other special products listed in the Ford Car Care Products Manual.

PROCEDURE "A" - Door Crack Repair (less than 51mm/2″ in length)

Refer to the Parts listing table later in this TSB - Use Parts Kit 1L3Z-15202W06-AA.

1. Follow the appropriate F-Series Workshop Manual procedure for removal of door trim panel, inner and outer belt moulding, door glass, flock run, and outer door handle and mirror.
2. Disconnect the battery.
3. Place the Steel Reinforcement (LH: XL3Z-1520255-AA or RH: XL3Z-1520254-AA) over the outside of the door sheet metal and mark with pen or pencil. This is for placement visual reference only.
4. Use an angle grinder or die grinder with a burr to open the crack area slightly and remove paint in the repair location.
5. Examine the area around the crack to verify the surface is flat and not recessed. Adjust the metal as necessary.
6. Cover the door opening of the cab with a welding blanket.
7. Weld the crack to a slight over-crown.
8. Right angle grind the welded area with an 80-grit disk and scuff the surrounding area with a Scotchbrite pad. Feather edge the repair, keeping it as small as possible.
9. Right angle grind the steel reinforcement with an 80-grit disk, sanding the back (mating surface) of the metal reinforcement for a good adhesive bond.

Copyright © 2001 Ford Motor Company

10. Move to the inside of the door and remove the cardboard retainer and two (2) Christmas tree pins from the corner of the door.
11. Sand the inside of the cracked outer door panel (area where steel reinforcement will be located). Use self-adhesive 80-grit sandpaper attached to a paint stick to get inside the door opening. Be sure to cover the end of the paint stick with the sandpaper. Then sand (window channel area) where the ABS reinforcement will fit in the door.
12. Blow out the door with clean compressed air.
13. Clean the ABS and steel reinforcements and sanded areas inside of the door with an Alcohol Wipe (ZC-4).
14. Attach the steel reinforcement to a paint stick using two-sided sticky tape.
15. Install the Motorcraft TA-3 Adhesive Cartridge into the Application Gun (501-040). Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive.
16. Apply Motorcraft TA-3 Adhesive to the Steel Reinforcement (LH: XL3Z-1520255-AA or RH: XL3Z-1520254-AA). Be sure all mating surfaces are fully covered.
17. Install the steel reinforcement between the inner and outer door panels. Install the lag bolts through the holes in the door inner panel (where Christmas tree pins were removed). Make sure of a tight even pressure fit on the reinforcement (Figure 2).
18. Check with a screwdriver to be sure the steel reinforcement is flush to the radius and bottom lines up with mark on the outside of the door.
19. Allow 20 minutes for the adhesive to reach handling strength. Work time is 5 to 8 minutes at 21°C (70°F). Full cure time is 4 hours.
20. Dry-fit the ABS reinforcement inside the door at window channel area (Figure 1).
21. Install Motorcraft TA-9 Adhesive Cartridge into the Application Gun (501-040). Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive.
22. Apply Motorcraft TA-9 Adhesive to the ABS Reinforcement (LH: 1L34-1520245-AA or RH: 1L34-1520244-AA). Be sure all mating surfaces are fully covered (Figures 3 and 4).
23. Seat plastic reinforcement firmly below window channel run. Check to be sure window run fits tightly into notch in ABS reinforcement (Figure 1).
24. Wipe off any excess adhesive from end of window run.
25. On the outside of the door, apply plastic filler to the repair area as needed. Sand the filler with 80-grit sandpaper, finish with 240-grit, then blow off the area with clean compressed air.
26. Clean with wax and grease remover and prep the area.
27. Scotchbrite the entire door and blend area.
28. Clean and prep for paint.
29. Mask the door area as necessary.
30. Mix and apply Ford-approved primer surfacer. Force dry the material at 60°C (140°F) panel temperature, using an oven or Infra-Red (IR) lighting.
31. Block sand the primer surfacer with 400-grit sandpaper and blow off with clean compressed air.
32. Mix and apply to blend Ford-approved basecoat material in the repair area. Allow material to flash off.
33. Mix and apply Ford-approved clearcoat to the entire door. Allow the material to flash off, then bake at 60°C (140°F) panel temperature per paint manufacturer's recommendations.
34. Unmask the vehicle.
35. Remove the lag bolts and reinstall cardboard reinforcement and Christmas tree pins, door glass, and glass channel flock run. Tighten outside mirror, outside door handle, door moulding, and door trim panel. Reconnect battery.

PROCEDURE "B" - Crack In Door Panel (longer than 51mm/2″ in length) Requires outer door panel replacement.

Refer to the Parts listing table later in this TSB - Use Parts Kit TA-7A.

NOTE
WHEN REPLACING THE OUTER DOOR PANEL ON 1997-2000 MODEL YEARS, THE WINDOW OUTER BELT MOULDING WILL ALSO REQUIRE REPLACEMENT. THE OLD STYLE OUTER BELT MOULDING CANNOT BE USED WITH THE NEW STYLE OUTER DOOR PANEL.

1. Follow the appropriate F-Series Workshop Manual procedure for the removal of the door trim panel, inner and outer belt moulding, exterior door handle and lock, window glass, mirror, wiring harness and flock run from the door.
2. Disconnect the battery.
3. Drill out the welds in the upper door frame (window opening).

NOTE
THE NEW OUTER DOOR PANEL WILL USE ADHESIVE IN PLACE OF THE ORIGINAL SPOT WELDS.

4. Mark the door hinges and using two (2) people, remove the door from the hinges. Leave the hinges on the pillar post.
5. Lay the door on a rack or sawhorse. Remove the door weatherstrip and leave the clips in place.
6. Use a screwdriver and a hammer to slightly lift the corner of the hem flange. Insert Tool 501-078/1 RT or Tool 501-078/2 in an air hammer and lift the hem flange around the entire door. 90 psi. air pressure is required for consistent tool operation.
7. Pop the welds around the window opening with a screwdriver.
8. Use a heat gun and heat the skin at the door impact bar area, to loosen the foam mastic holding the door skin, lift and remove it.
9. Use a small angle grinder to clean paint, adhesive, and corrosion from the mating surfaces of the inner panel, then lightly scuff the new door panel mating flange with a red Scotchbrite pad.
10. Blow off the area with compressed air and apply masking tape (1/2″ wide) to the area that will be covered by the adhesive.
11. Mix and brush on Ford-approved epoxy or self-etching primer to all unpainted flange areas inside the door.
12. Cure the primer at 140°F panel temperature per paint manufacturers label instructions.
13. Remove the masking tape from the door mating flange and dry fit the door skin.

NOTE
ABS REINFORCEMENT BLOCKS ARE NOT REQUIRED WHEN REPLACING OUTER DOOR PANEL.

14. Install the Motorcraft TA-1 Adhesive Cartridge into the Application Gun 501-040. Squeeze out some material to purge the cartridge. Install the tip on the cartridge and squeeze out some material to make sure of even mixing of the adhesive components.
15. Remove masking tape and apply the Motorcraft TA-1 Adhesive to the mating flange of the inner door panel.
16. Fit and clamp the outer door panel to the inner panel. Work time is 45 minutes at 21°C (70°F). Handling strength in 90 minutes. Full cure time is 8 hours.
17. Use tool 501-080 to close down the entire hem flange. Start to close the hem by putting the flange in the middle of the tool and hit the tool. Continue around the door closing down the hem flange.
18. Apply Motorcraft TA-2 Urethane Seam Sealer around the door hem flange to give the repair a finished look
19. Remove any residual sealer with adhesive remover. Sealer is paintable in 10 minutes and within 24 hours
20. Install the Motorcraft TA-4 Adhesive Cartridge into the (caulk type) Application Gun. Squeeze out some material to purge the cartridge. Install the tip on the cartridge and apply between the intrusion bar and the door skin. Apply mastic foam between the intrusion bar and the door skin.
21. Blow off the door with compressed air to remove debris.
22. Use wax and grease remover to clean the door for paint.

23. Mask the inside of the door for applying paint.
24. Mix and apply Approved Basecoat Color to the door and allow material to flash off.
25. Mix and apply Clearcoat to the door. Following manufacturers recommended baking time at 140°F panel temperature
26. Demask the door.
27. Reinstall door weather stripping.
28. Reinstall the door, and wiring harness.
29. Sand the door and fender with 360 grit sand paper prepping for a blend. Blow off the panels with clean compressed air. Clean the panels with wax and remover.
30. Mask off and apply stone abrasion material to the bottom of the door.
31. Mix and apply Ford Approved Basecoat material to the repair area and let flash.
32. Mix and apply Ford-approved two-tone basecoat color as necessary and let flash.
33. Mix and apply Ford Approved Clearcoat to the repair area and bake at 140°F panel temperature at manufacturer's recommended bake time.
34. Demask the vehicle.
35. Reinstall mirror, exterior door handle, window glass, flock run, door trim panel, and stripes and moulding as necessary.

Procedure "A" Parts and Materials List

KIT 1L3Z-15202W06-AA INCLUDES	
PART	**PART NUMBER**
RH ABS Reinforcement	1L34-1520244-AA
RH Steel Reinforcement	XL3Z-1520254-AA
LH ABS Reinforcement	1L34-1520245-AA
LH Steel Reinforcement	XL3Z-1520255-AA
Motorcraft Metal Patch Panel Adhesive	TA-3
Motorcraft Isopropanol Cleaning Kit	ZC-4
Motorcraft Plastic Bonding Adhesive	TA-9
Mixing Tips (quantity - 4)	N/A
Lag Bolts - 1/4″x4″ (quantity - 2)	N/A

Procedure "B" Parts and Materials List

KIT TA-7A (Motorcraft) INCLUDES	
PART	**PART NUMBER**
Motorcraft Metal Bonding Adhesive	TA-1
Motorcraft Repair Urethane Adhesive	TA-2
Motorcraft Flexible Foam Repair	TA-4

PROCEDURES "A" AND "B" TOOLS

SPECIAL TOOLS - (Tools are direct ship from O.T.C.)
501-078/1 RH Hem Opening Tool 501-078/2 LH Hem Opening Tool 501-080 Hem Closing Tool 501-040 Adhesive Gun (Adhesive Gun not included in kits. This is a commonly used commercially available gun. Must be ordered separately from Rotunda.)

NOTE

THE APPLICATION GUN IS NOT INCLUDED IN KITS. THIS IS A COMMONLY USED, COMMERCIALLY AVAILABLE GUN. IT MUST BE ORDERED SEPARATELY FROM ROTUNDA (501-040).

LABOR CLAIMING EXAMPLE

2000 F-150 Regular Cab - Dealer is repairing door crack on one (1) front door. Dealer would claim operations as follows:

- 011802AA (1.3) - Time to repair doorcrack
- 011802AAP (2.0) - Additional time to paint
- P101 (0.5) - Basic paint operation (once per visit) or P102 (0.7) - If vehicle is equipped with two tone paint

Material allowance is calculated only on the paint labor time (P101 or P102, 011802AAP). Total the paint labor time, multiply time x 14.75 for the total material allowance. Use (PAINT) as the causal part for claiming material allowance.

- Part #: "Paint"
- QTY.: 2.5
- EA. Amt.: $14.75
- Total = 36.88

This repair would pay 3.8 hours (1.3 + 2.0 + 0.5 = 3.8) for the total job plus 2.5 hours (2.0 + 0.5 = 2.5) for the material allowance.

PART NUMBER	PART NAME
1L3Z-15202W06-AA	Kit - Door Reinforcement
TA-7A (Motorcraft)	Kit - Sheet Metal Bonding
Obtain Locally	Stone Abrasion Material

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 01-2-6
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
011802AAP	Additional Time To Paint For Operation AA	2.0 Hrs.
011802AA	Perform Door Crack Repair Procedure One Front Door	1.3 Hrs.
011802ABP	Additional Time To Paint For Operation AB	4.0 Hrs.
011802AB	Perform Door Crack Repair Procedure Both Front Doors	2.4 Hrs.
011802BAP	Additional Time To Paint For Operation BA	2.0 Hrs.
011802BA	Replace Outer Door Panel One Door	2.8 Hrs.
011802BBP	Additional Time To Paint For Operation BB	4.0 Hrs.
011802BB	Replace Outer Door Panel Both Doors	5.4 Hrs.
011802P101	(One Color) Can Be Claimed Once Per Visit Even If Multiple Panels Are Involved	0.5 Hr.
011802P102	(Two-Tone) Can Be Claimed Once Per Visit Even If Multiple Panels Are Involved	0.7 Hr.

DEALER CODING

BASIC PART NO.	CONDITION CODE
1520200	01

OASIS CODES: 106000, 111000

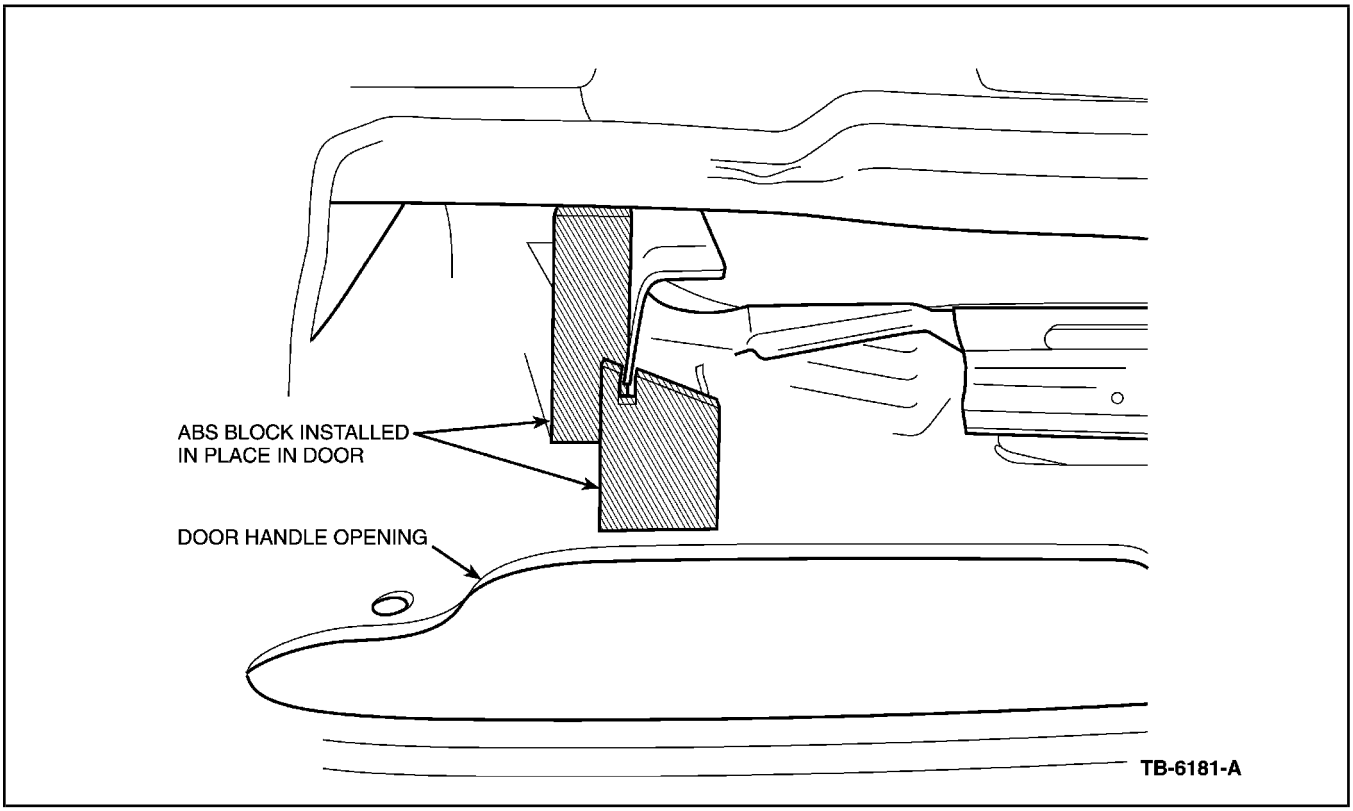


Figure 1 - Article 01-18-2

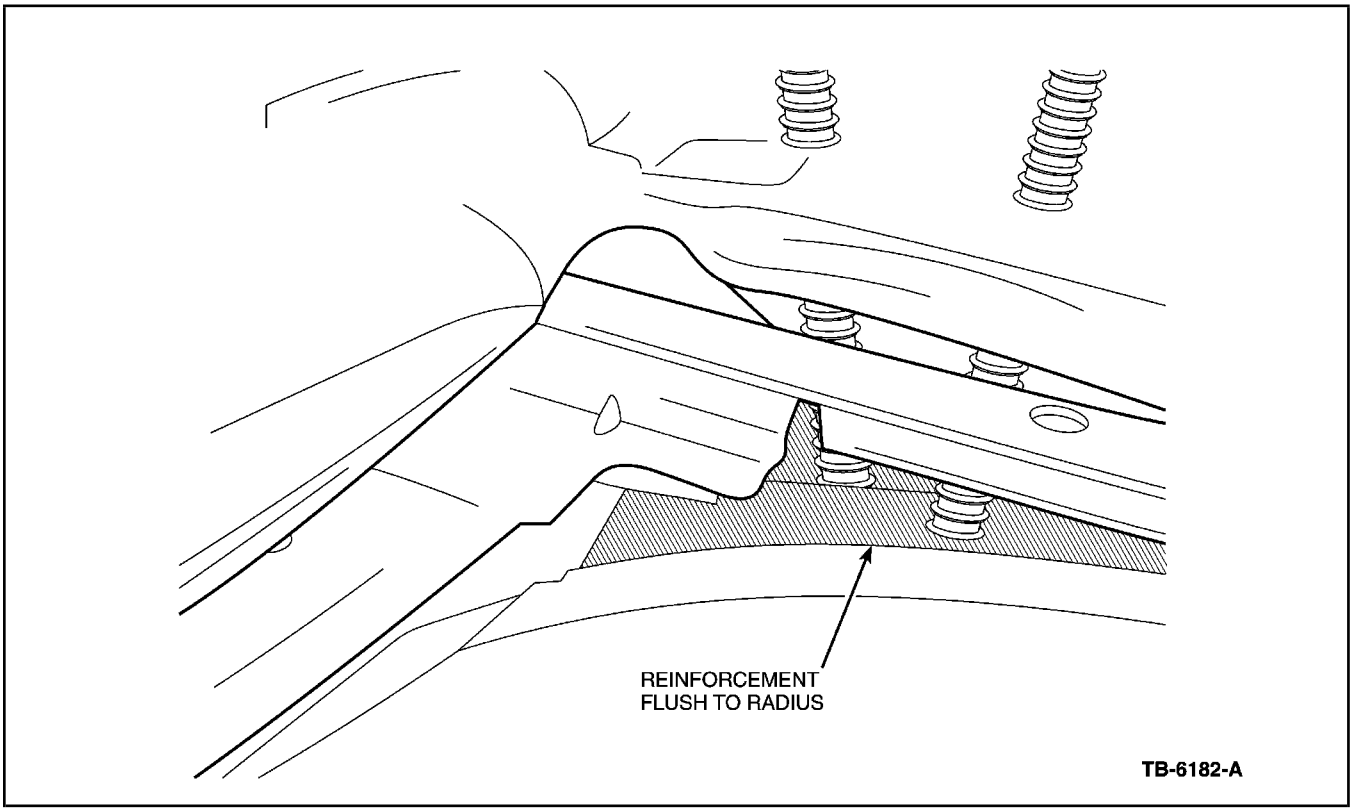


Figure 2 - Article 01-18-2

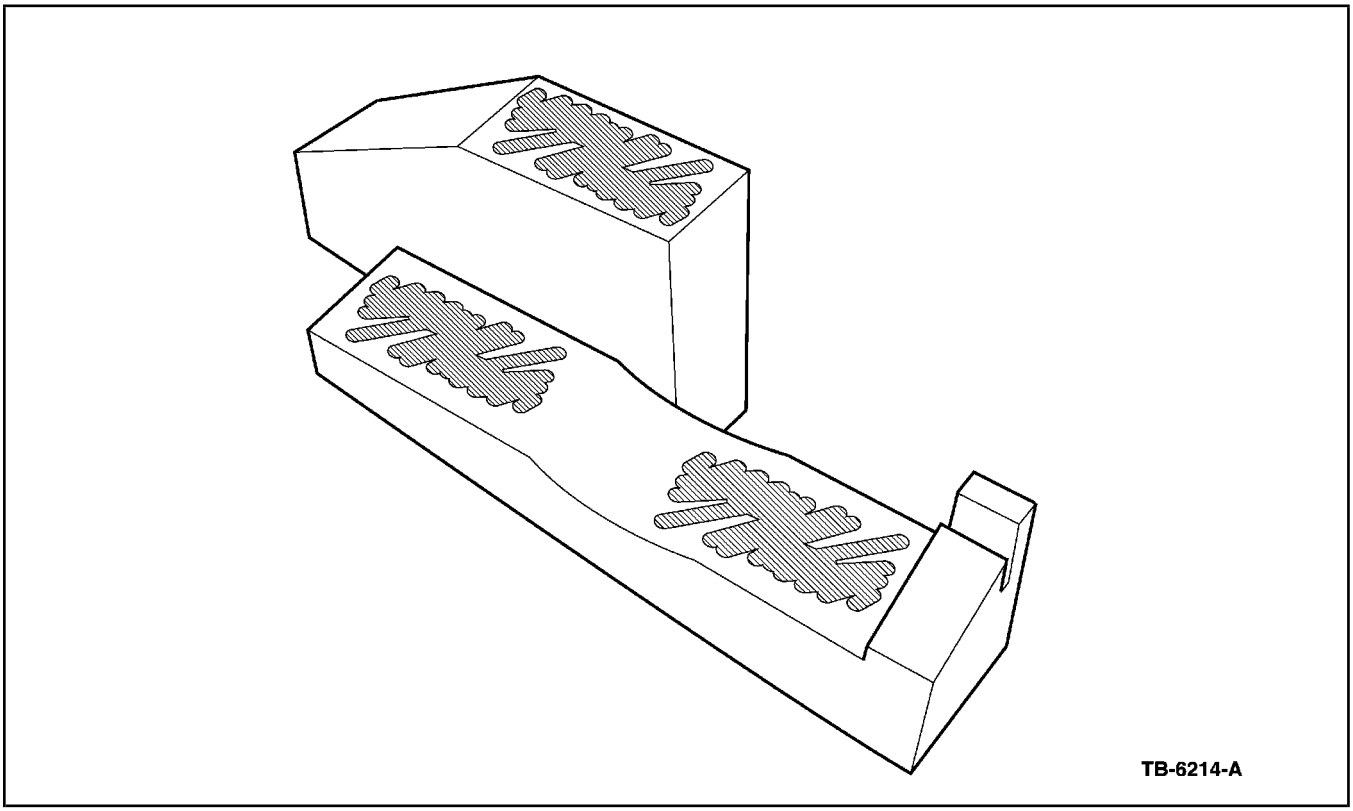
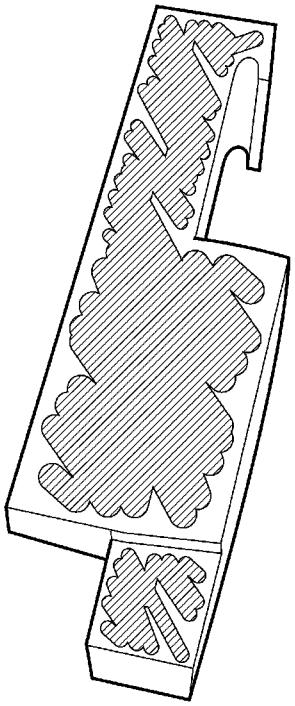


Figure 3 - Article 01-18-2



TB-6215-A

Figure 4 - Article 01-18-2